

Set of rules to solve a problem.

Algorithm: Steps to solve a problem finite steps
ETA 21/12/20

How to approach a problem?

→ each task in algo or each algo should complete in finite time.

Sum of two input

(P.S) → i/p → 100, 200
→ sum?

→ distance = 20 km, time = 2 hrs
speed? $s = d/t$

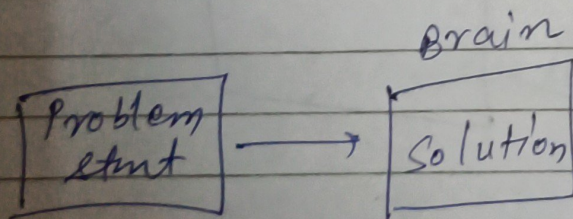
(A) Understand the problem.

(B) Input values.

(C) create logic / Algorithms finding

How?

to understand



Problem stmt → add 2 no.

Pseudocode

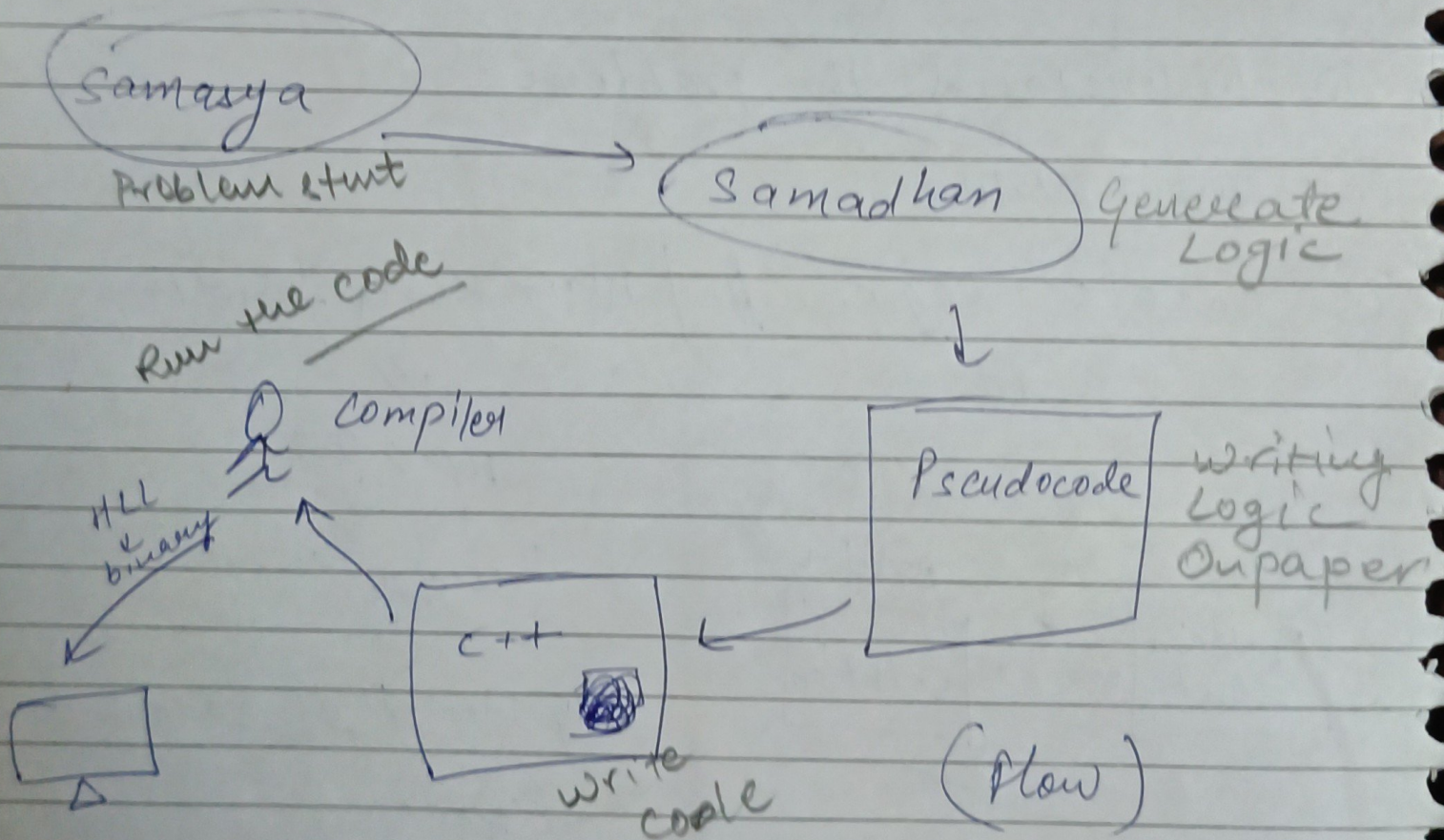
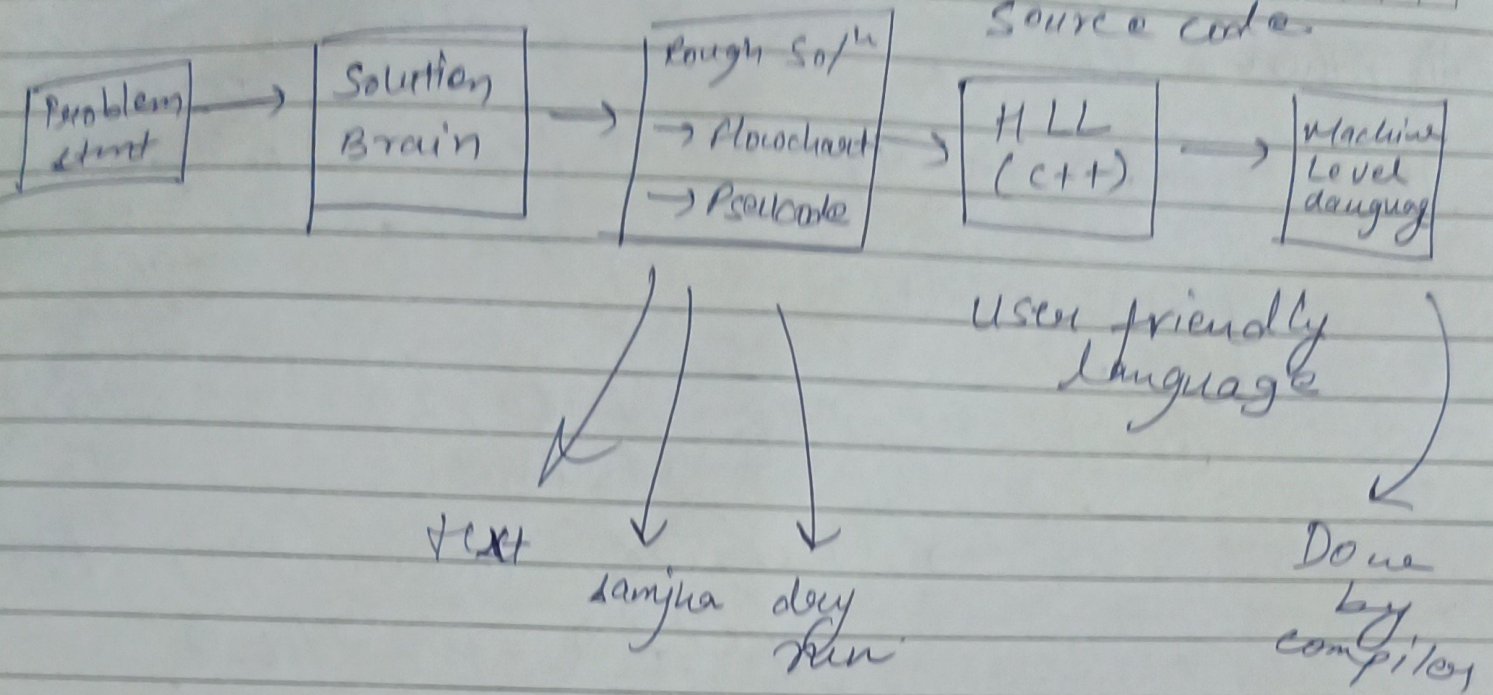
1) Take 2 i/p.

2) $sum = a + b$

3) print sum

Rough soln
→ Flowchart
→ Pseudocode

text
→ samjha
→ dry run



Low level lang.
Middle " "
High " "

Homework
use of pointers
can make as play
with memory by code

Prime no.

 $n = 17$

n

 $x = 2$ to $n-1$ divide $n \% x == 0$ Non Prime
return
loop ends.

x)

 $17/2 \rightarrow \text{rem} \neq 0$ $17/3 \rightarrow \text{rem} \neq 0$ $17/4 \rightarrow \text{rem} \neq 0$

⋮

 $17/16 \rightarrow \text{rem} \neq 0$

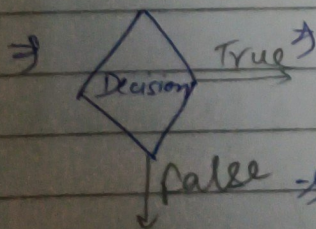
17 is prime Prime.

code is given using Logic (C++)

↓ compiler

Machine
level
languageFlowcharts: Diagrammatic repr. of an algo or process

components of flowcharts.

1) Starting / ending $\rightarrow$ start/end or start/endi/p / o/p
block

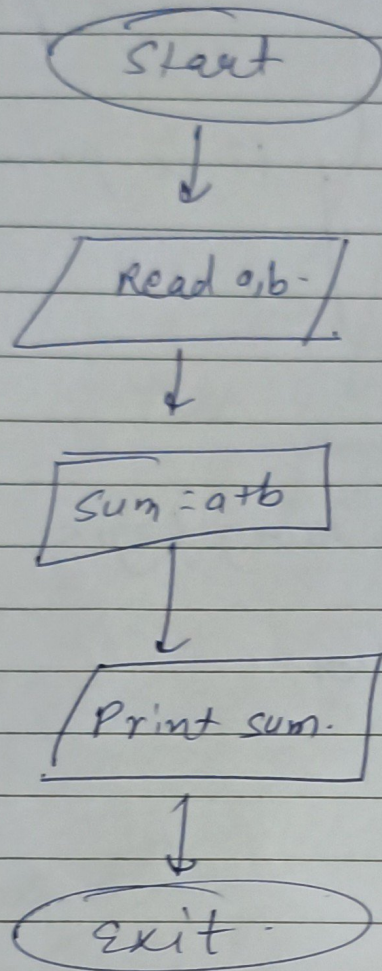
Process

Calculation / Initialisation /

Declaration

 $\rightarrow$ Arrows / connector

1) Print sum of a and b

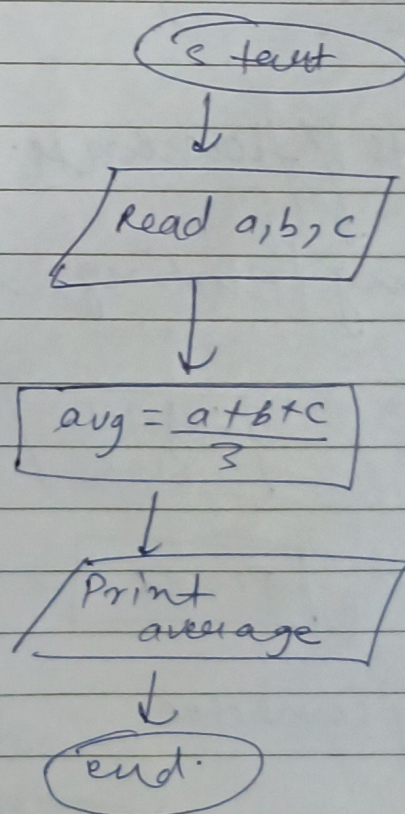


2) Average of a, b, c.

1) Input a, b, c

2)
$$\text{Average} = \frac{a+b+c}{3}$$

3) Print average



Flowcharts and Pseudocodes

Diagram to represent solutions of problems.

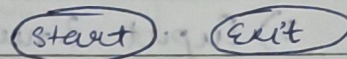
Divide into small parts



logically arrange

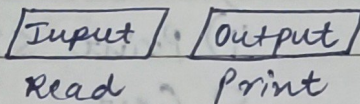
Components of Flowcharts:

1) Start / Exit



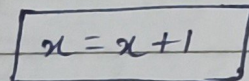
Oval

2) Input / output



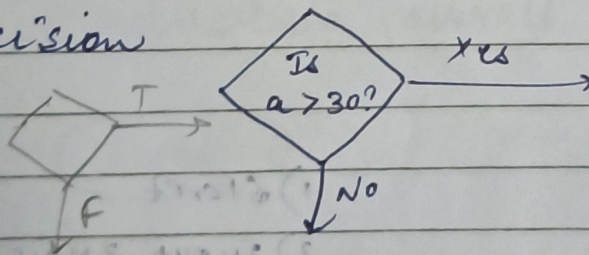
Parallelogram

3) Process



Rectangle

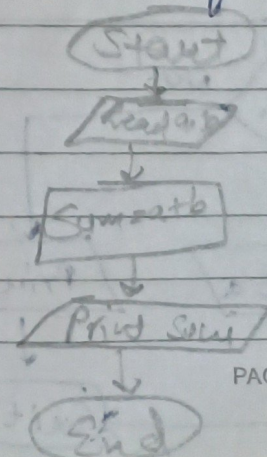
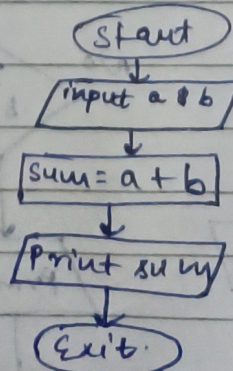
4) Decision



Diamond Shape
(Checks if something is true or false)

↓
Arrows show flow
connectors b/w different shapes of diagram

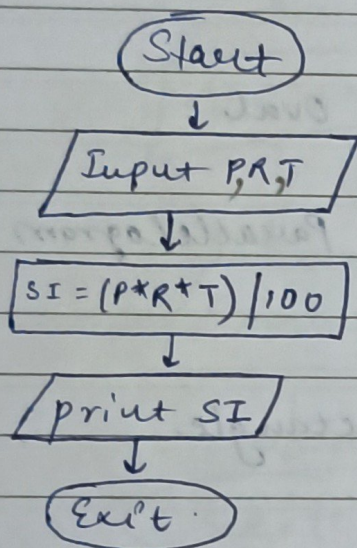
Ex) Sum of a and b



Pseudocode $\rightarrow$

1. Start
2. Input numbers a and b
3. Calculate $sum = a + b$
4. Print Sum
5. Exit

Ex) Calculate Simple Interest



1) Start

2) Input 3 numbers, P, R & T

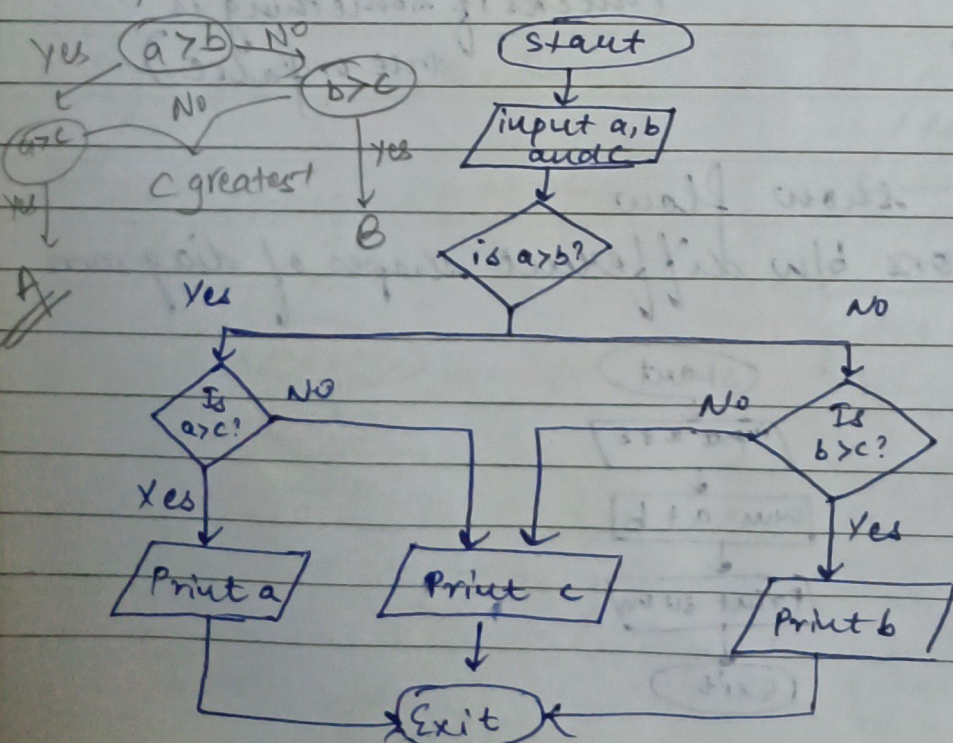
3) Calculate $SI = (P * R * T) / 100$

4) Print SI

5) Exit

Ex) Find max of 3 numbers.

a, b, c



1) Start

2) Input 3 no.s a, b and c

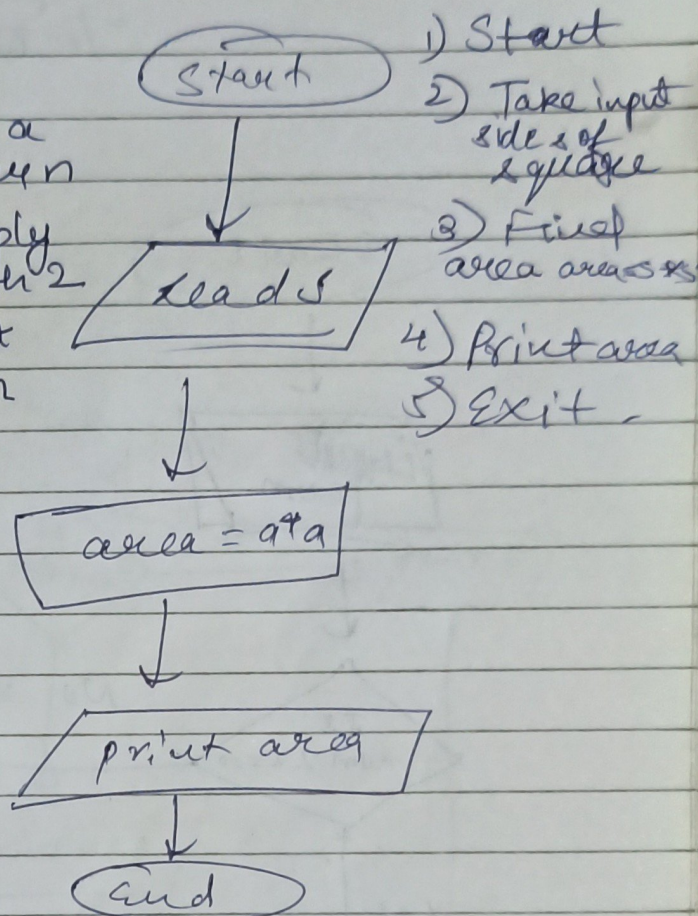
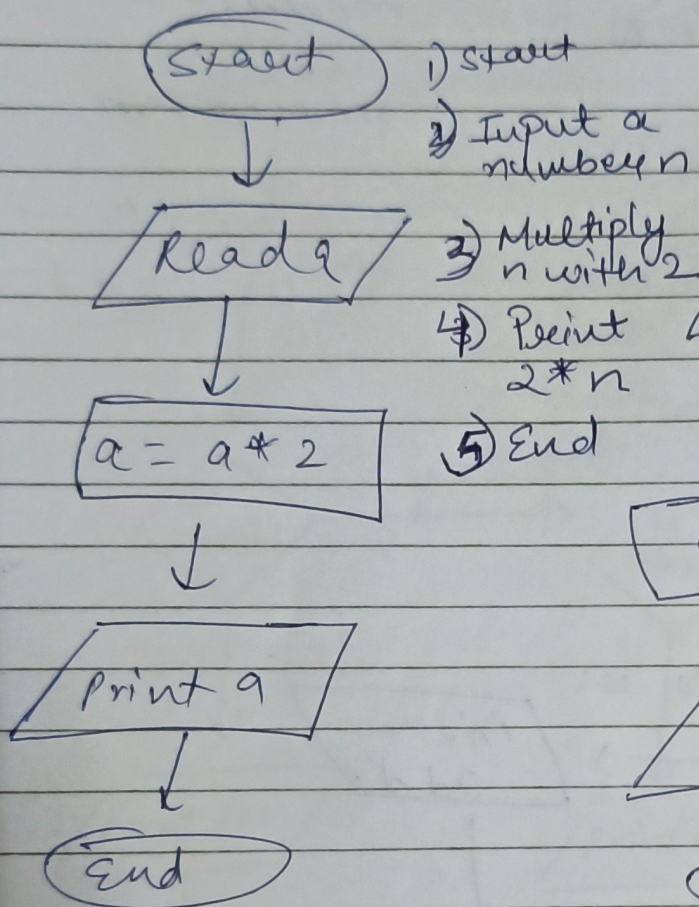
3) If $a > b$ do
 if $a > c$
 Print a
 else
 Print c

else
 if $b > c$
 Print b
 else
 Print c

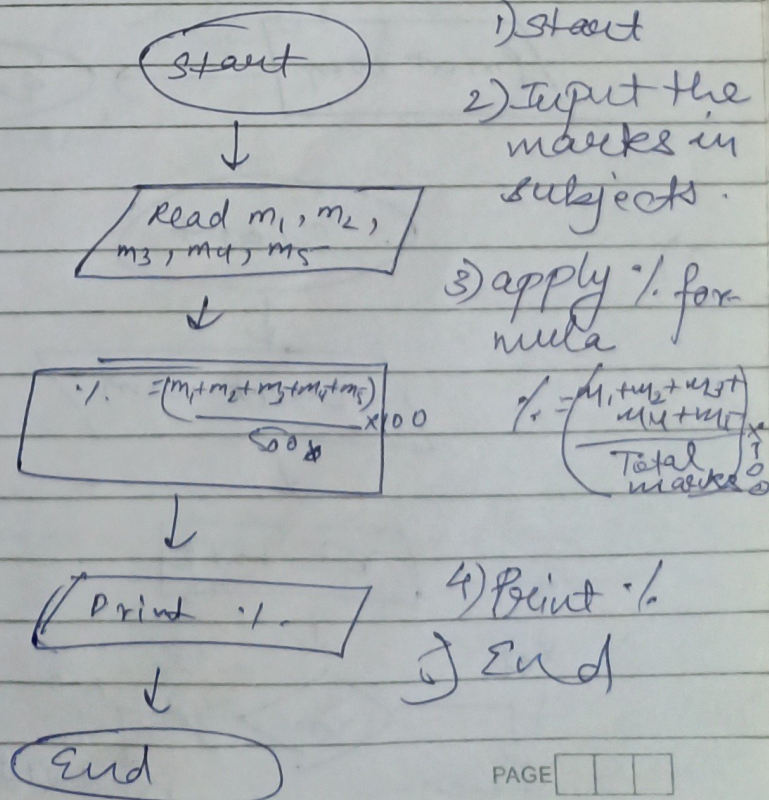
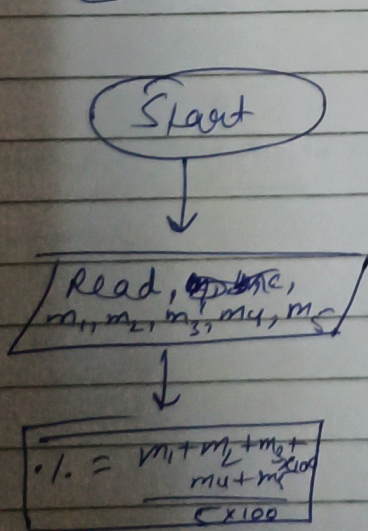
4) Exit

3) Twice of a number

4) Area of square



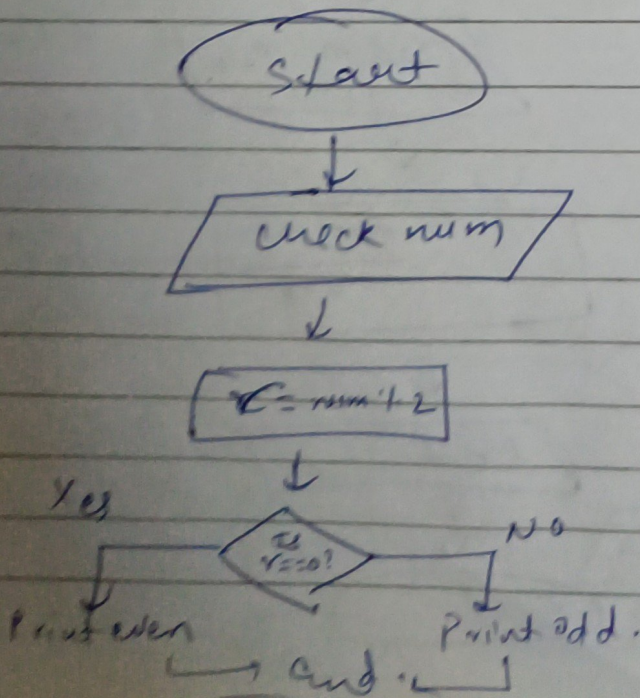
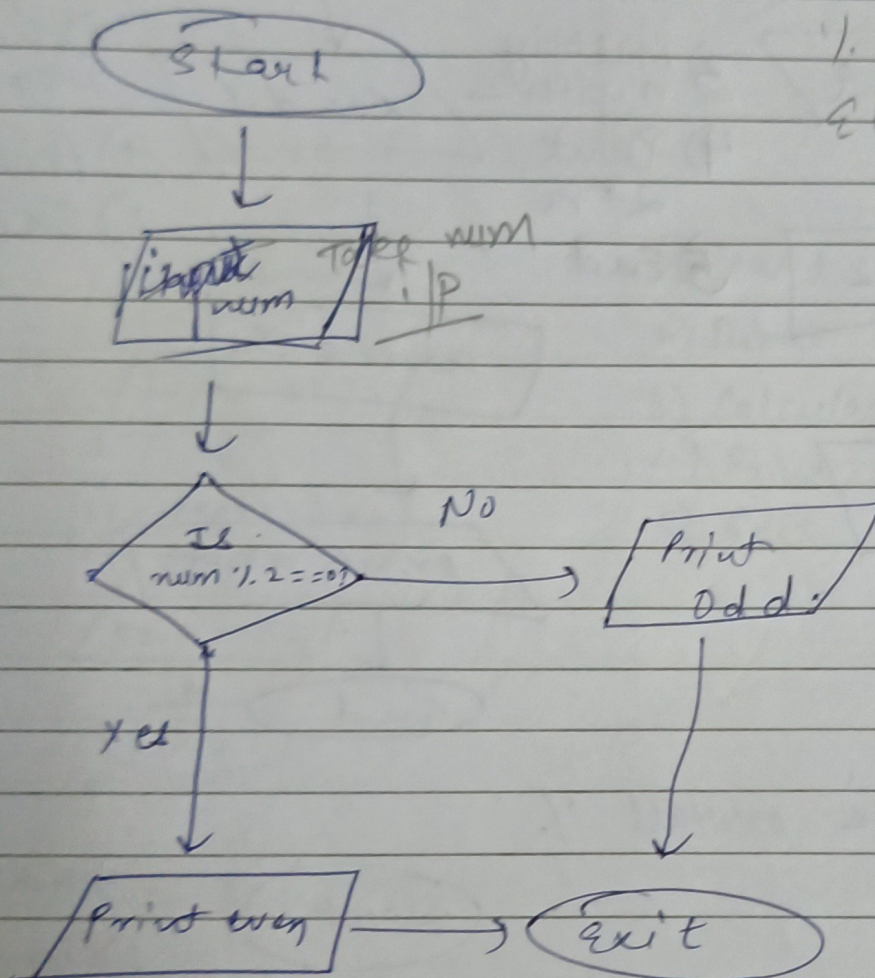
4) Calculate overall %



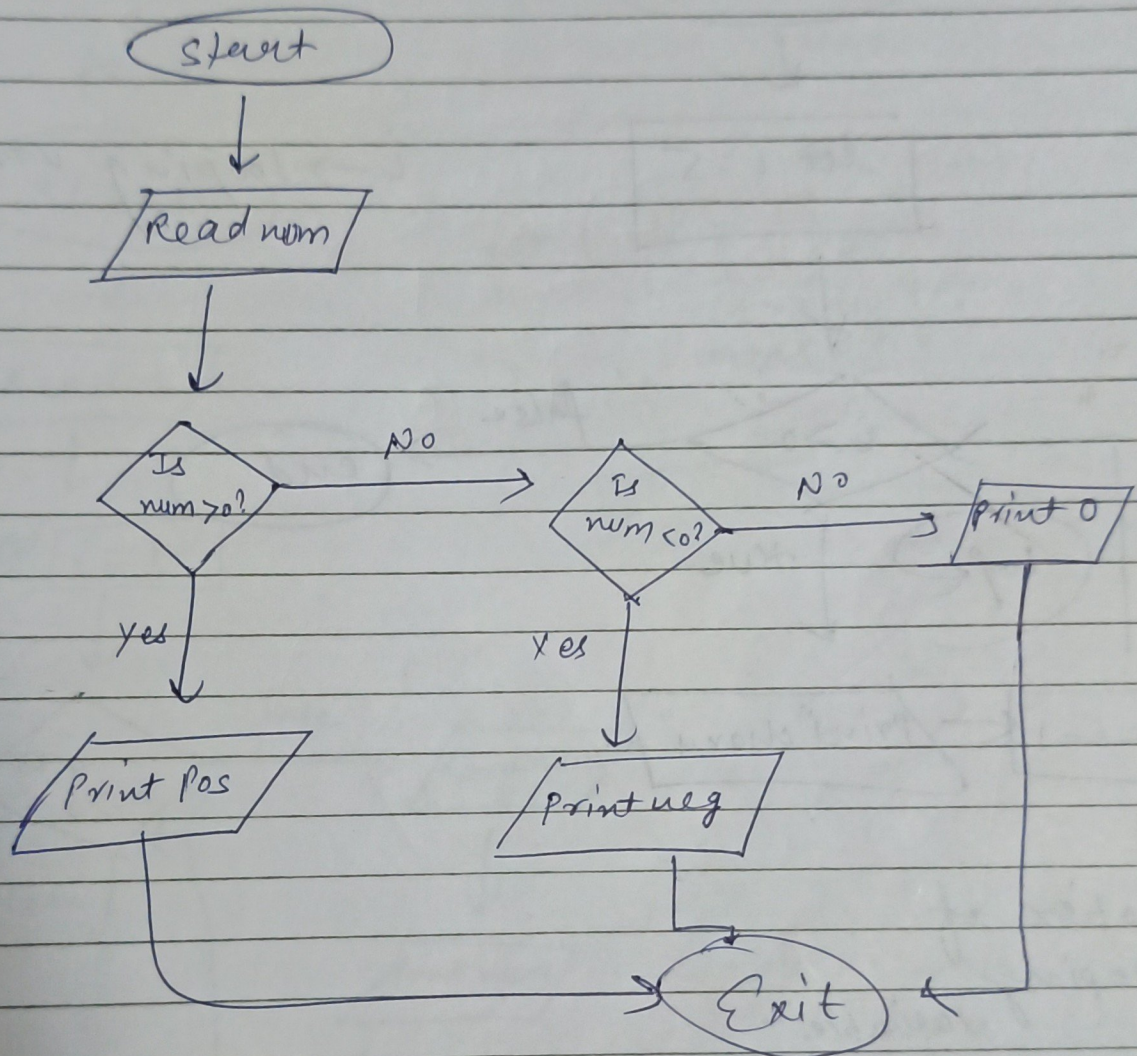
⑤ Check num is even or odd. Coded

$\text{num} \% 2$ modulous
↓
returning rem

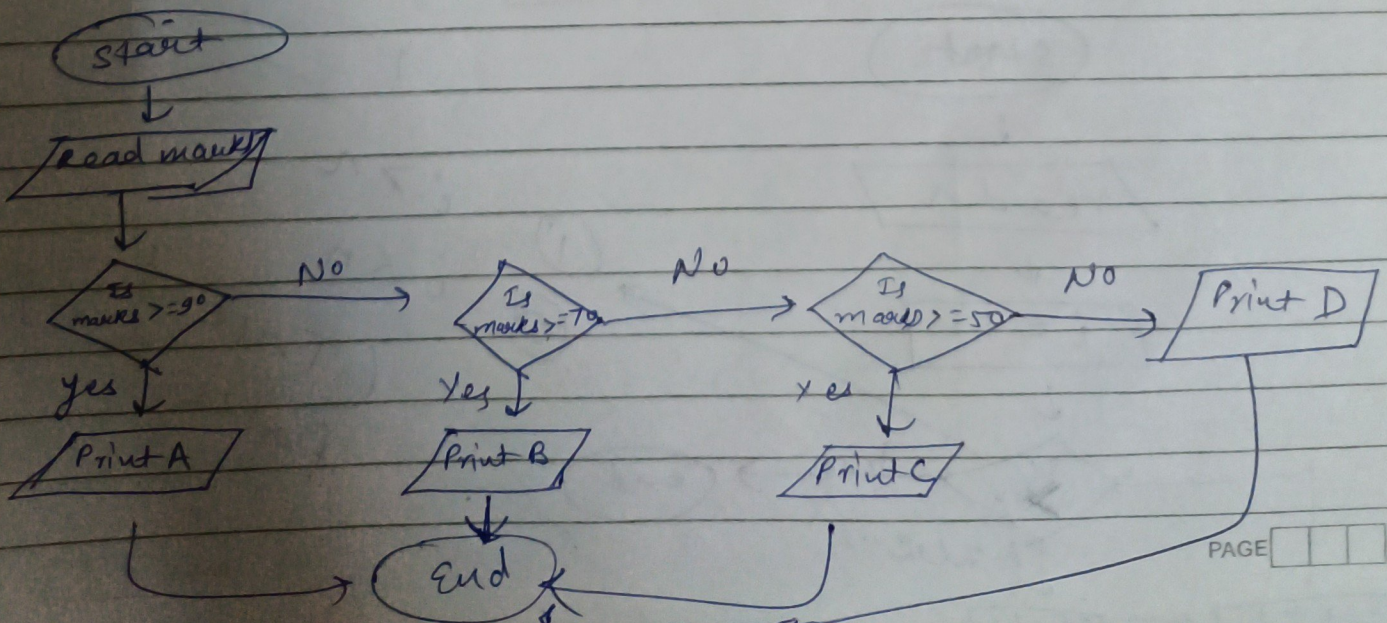
$\text{num} \% 2 == 0 \rightarrow \text{Even}$
else odd.



5) Check +ve, -ve or 0 - Coded



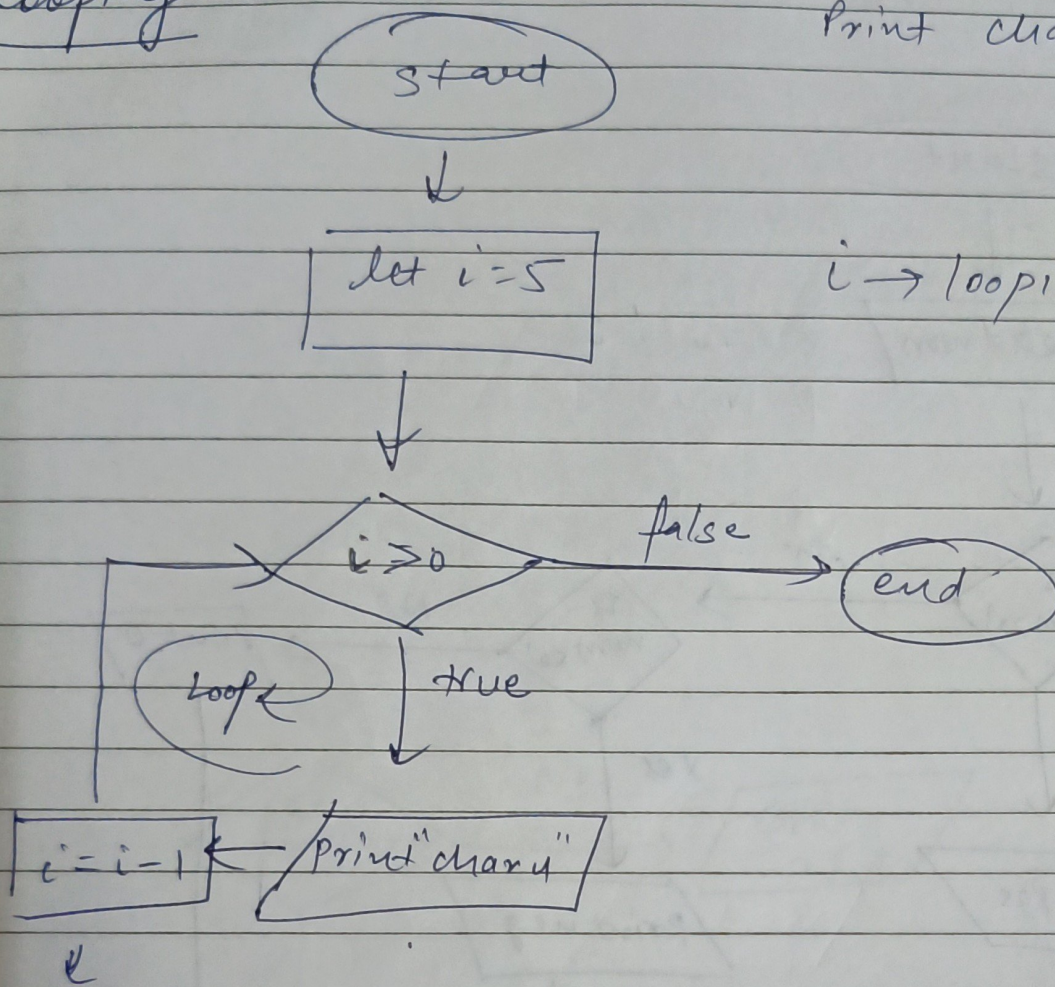
Student and Grade Flowchart



Looping

Print char 5 times.

$i \rightarrow$ looping variable.

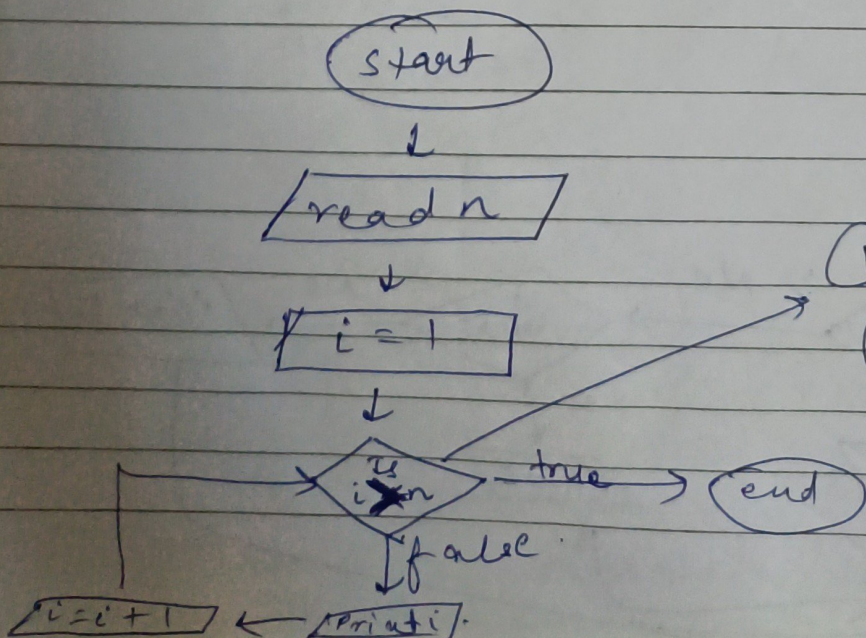


updaton of
looping
variable -

Print counting from 1 to n

```

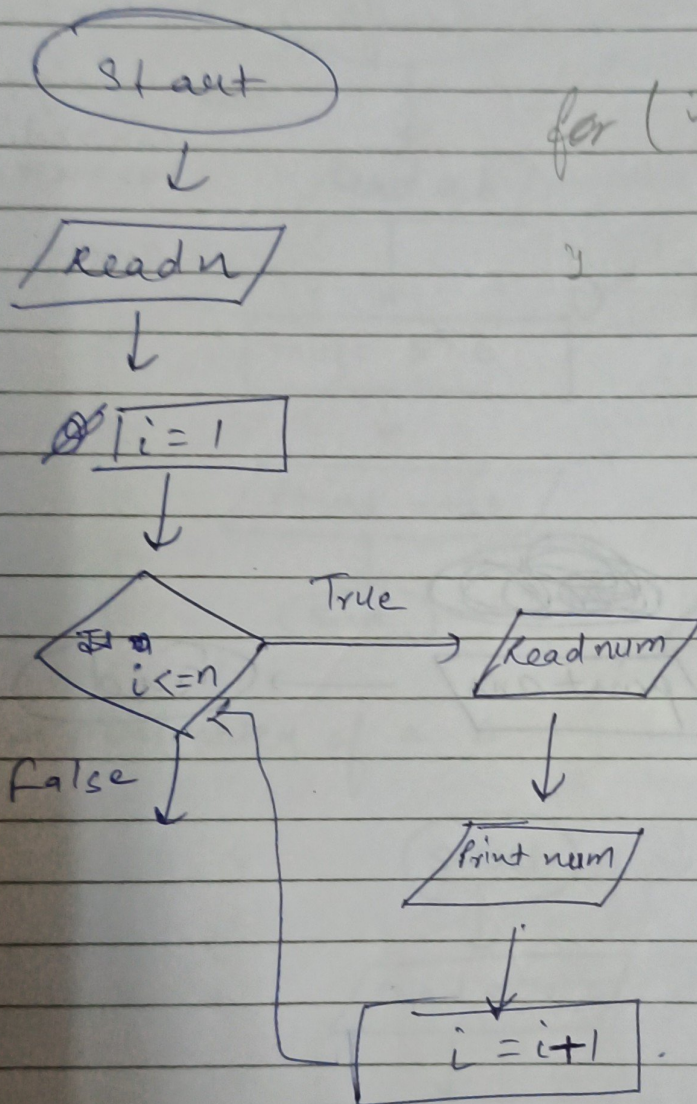
for(i=0; i<n; i++){
    cout << i+1;
}
    
```



$i > n$
 $i \leq n$

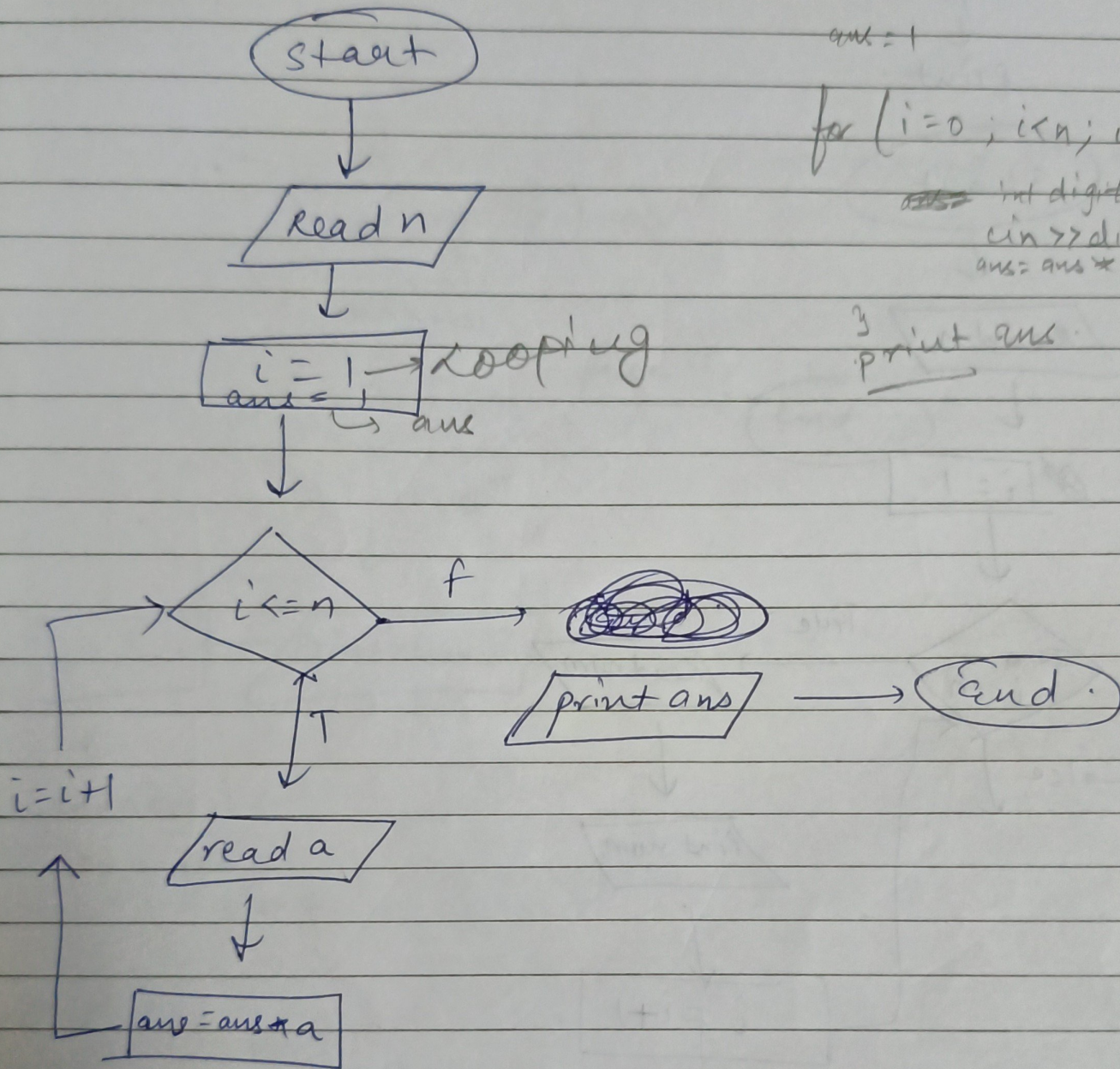
(True & false swapped)

Q) N times
↳ i/p
↓
print.

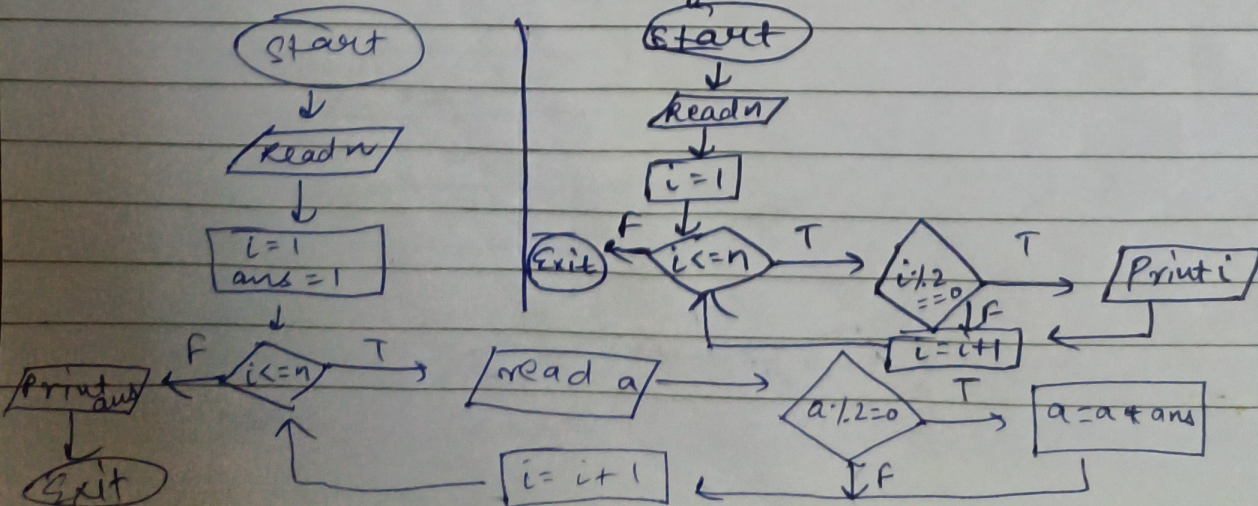


```
for (i = 0; i < n; i++) {  
    int digit;  
    cin >> digit;  
    cout << digit;
```

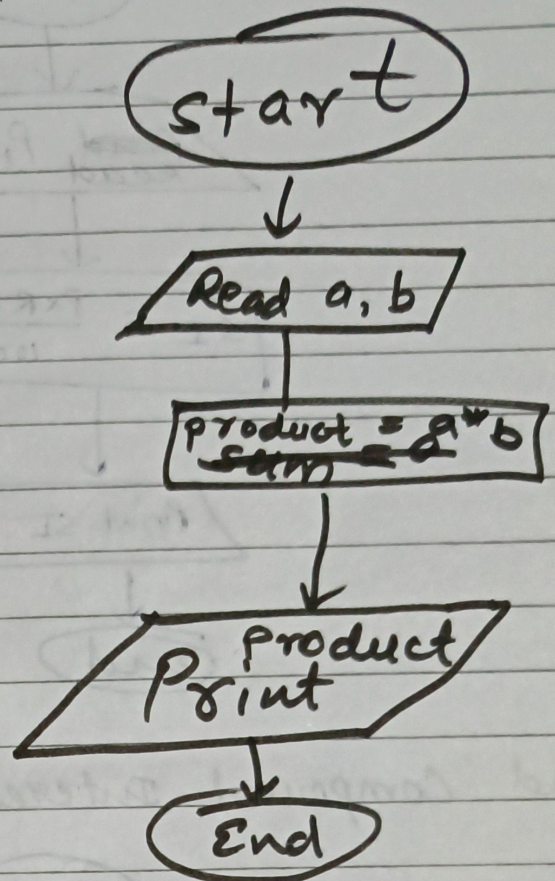
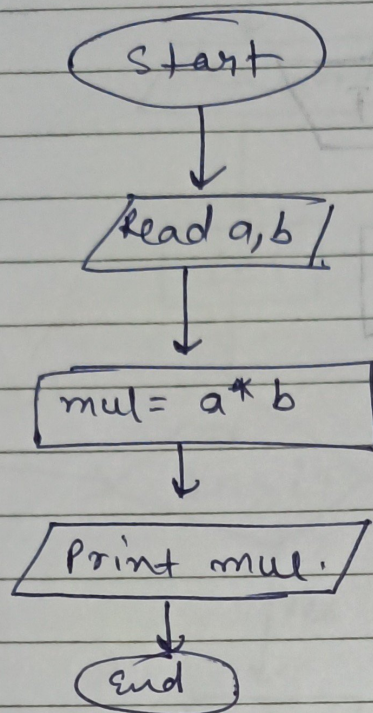

Multiply n numbers from user.



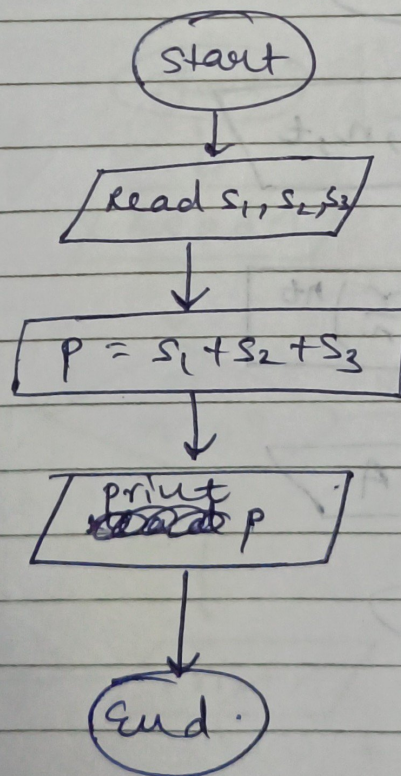
Print 1 to n; only even



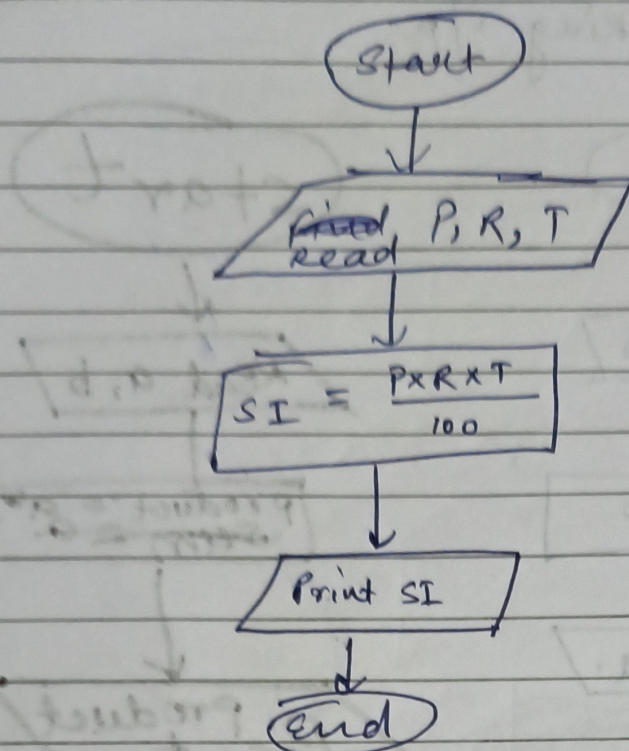
1) Multiply 2 no.s by taking i/p



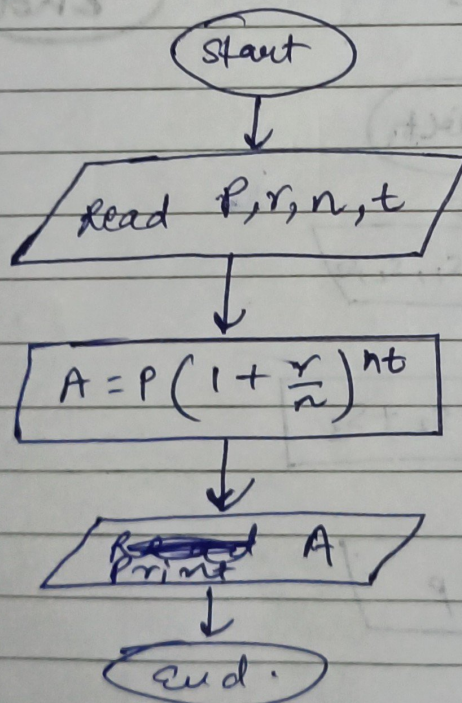
2) Find perimeter of a Δ



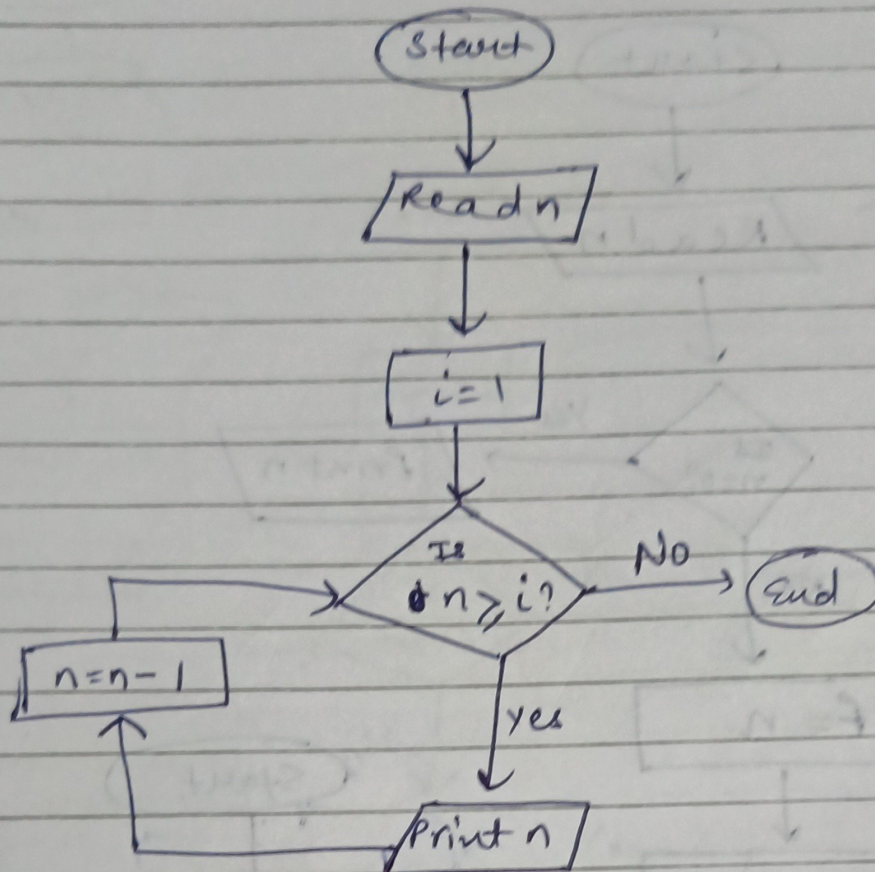
3) Find simple interest.



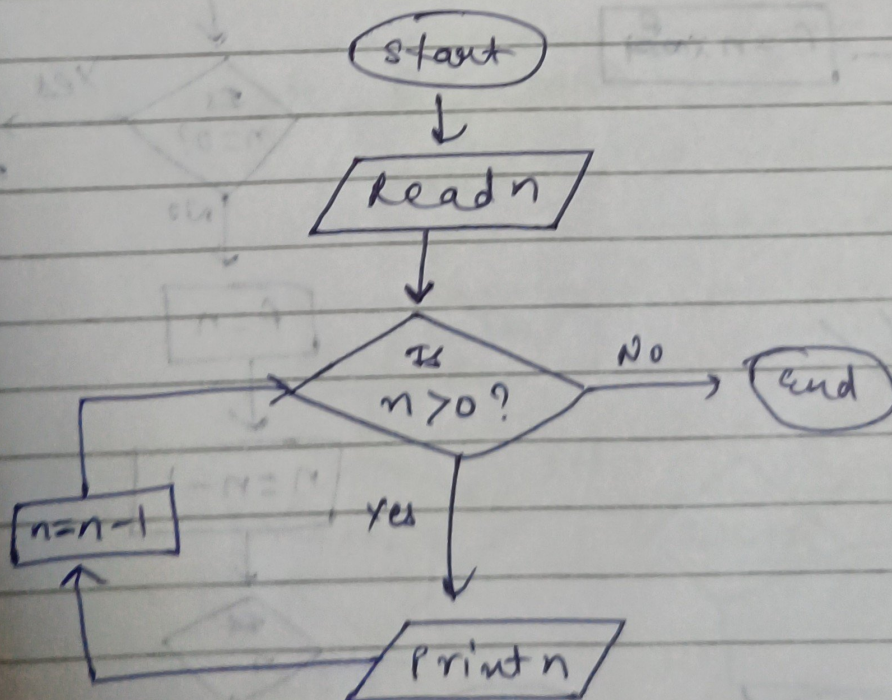
4) Find Compound Interest.



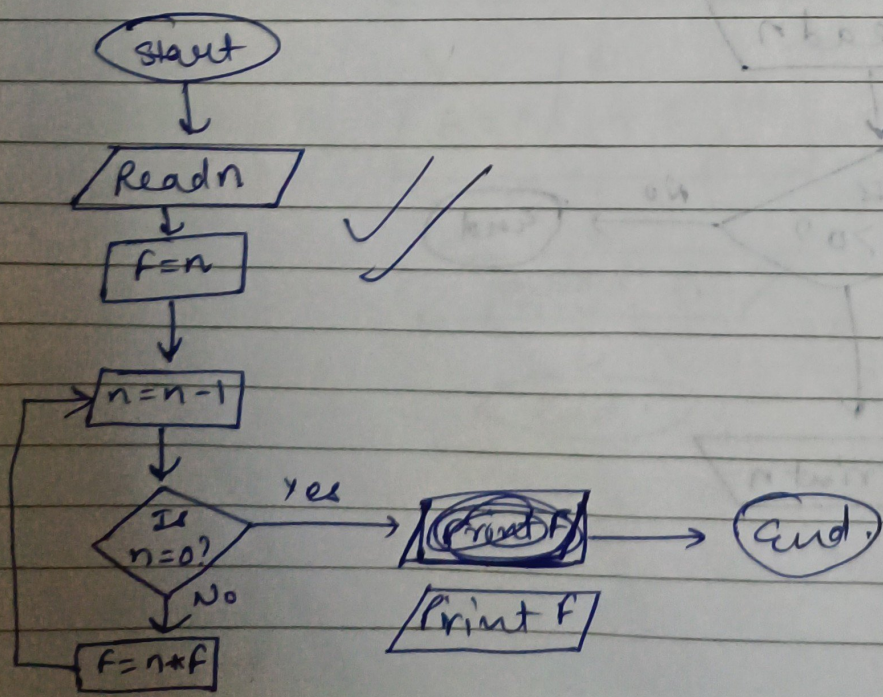
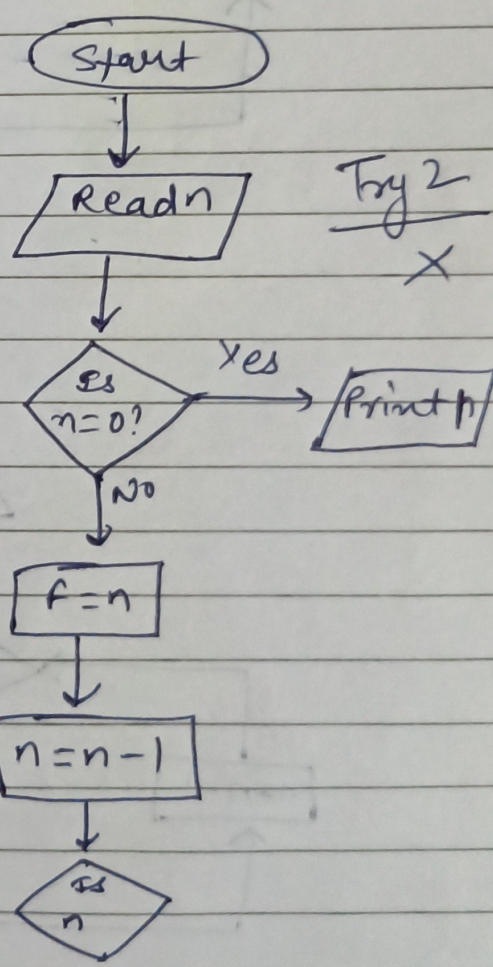
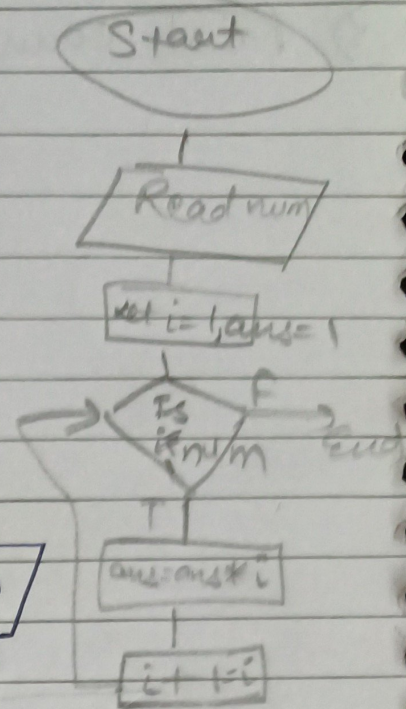
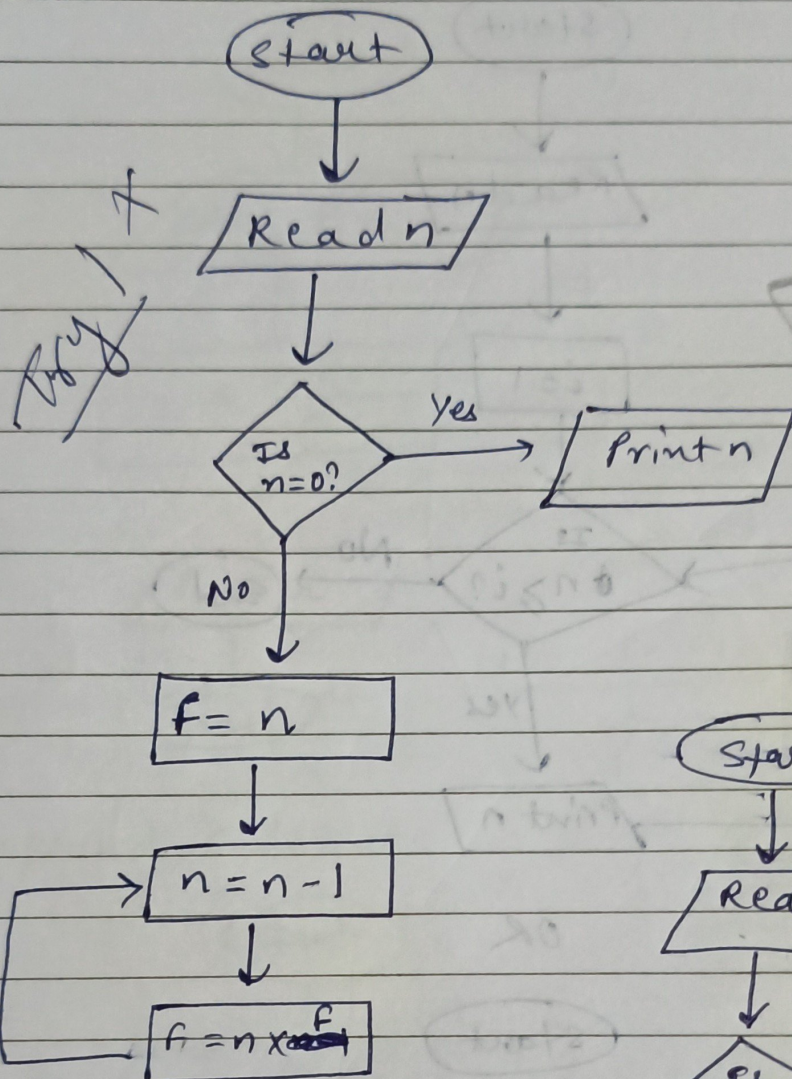
3) Print counting from N to 1



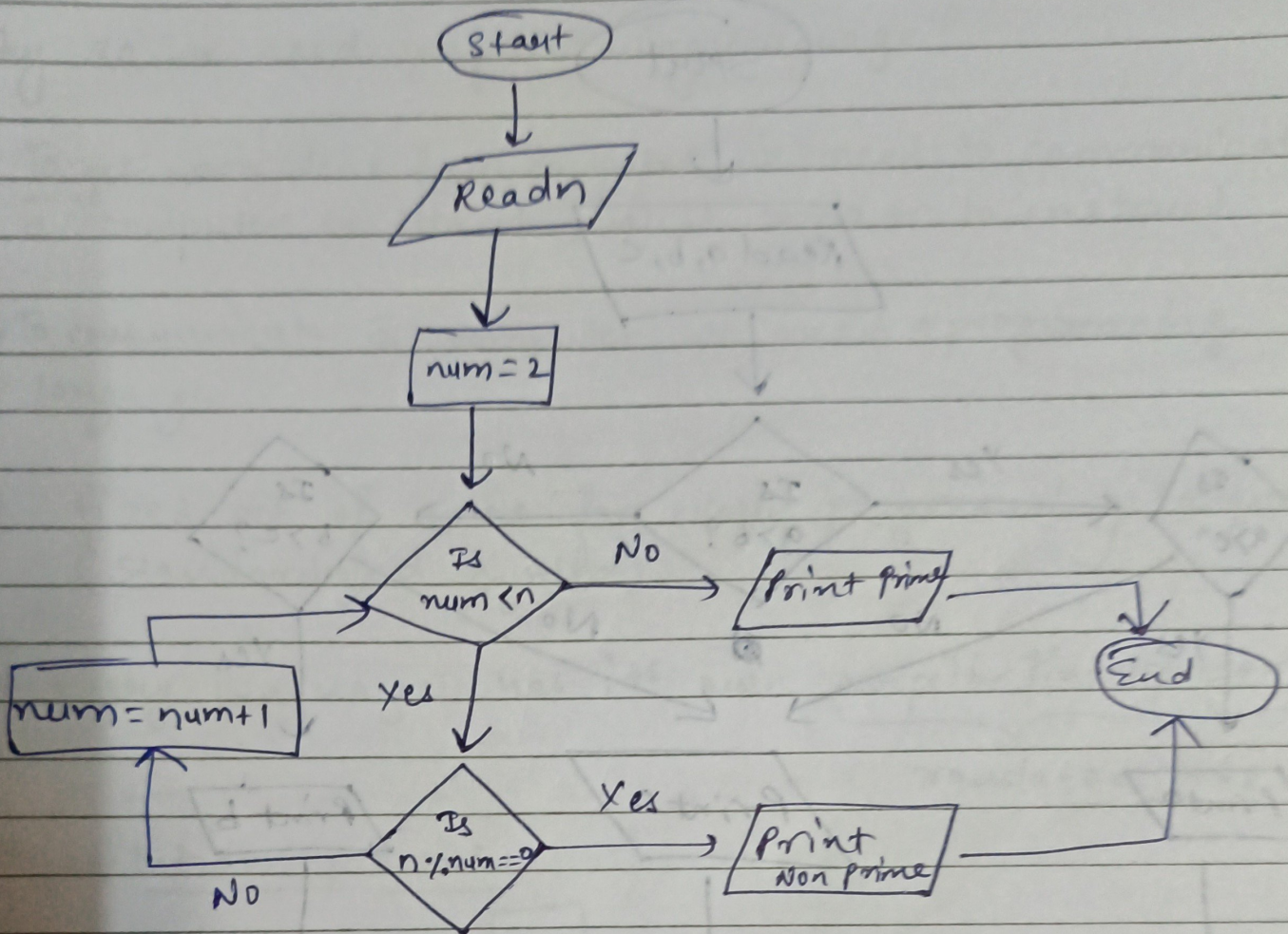
OR



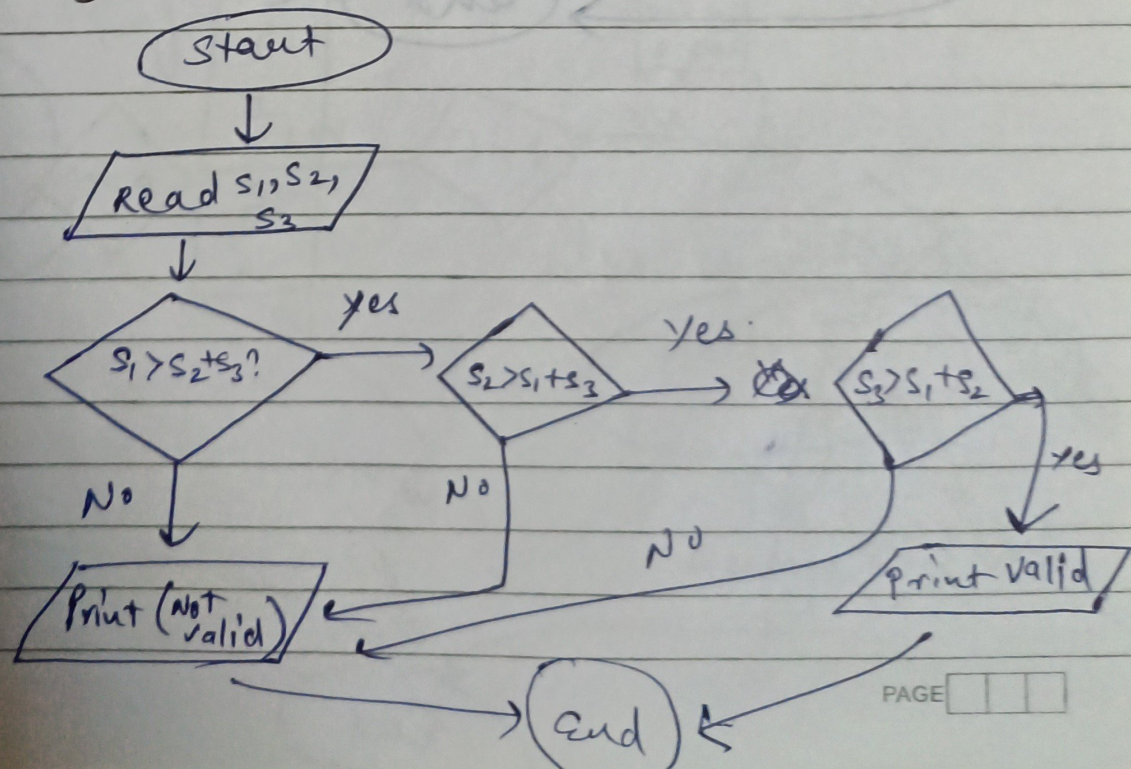
Q) Find factorial of a no.



1) Check if a number is prime or not.



2) Valid Triangle Or not



Q Print max of 3 numbers.

